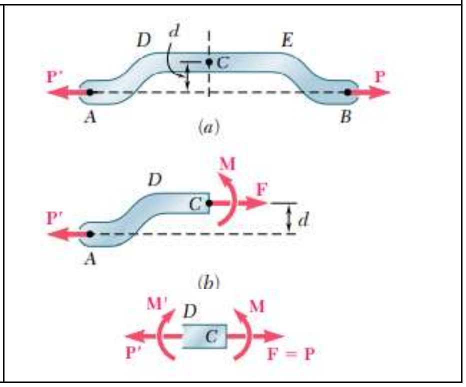
Bending due to eccentric axial loading in the plane of symmetry

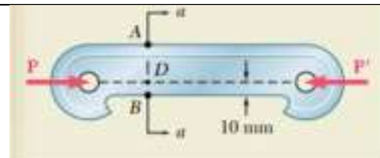
It was shown earlier while discussing the stresses in axially loaded members that the axial normal stress produced will be uniform only if the force acts through the centroid of the cross section. Such loads are called centric loads. If it does not, it is called eccentric load. First, let us consider eccentric axial force acting in the plane of symmetry of the cross section.

In such cases, the force should be transferred to the centroid of the cross-section. It can be seen that the eccentric axial load also produces a bending moment as shown. In such cases, we can find out the stresses separately for the axial force and the pure bending moment and using the principle of superposition the net stress can be obtained at any point in the cross-section.



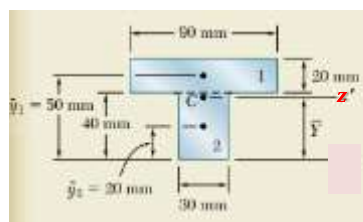
Example 1

Knowing that for the cast iron link shown the allowable stresses are $30\ MPa$ in tension and $120\ MPa$ in compression, determine the maximum force P that can be applied.



	A_i mm^2	y_i mm	$A_i y_i$
1.	$20 \times 90 = 1800$	50	90000
2.	$40 \times 30 = 1200$	20	24000
	$\Sigma A_i = 3000$		$\Sigma A_i y_i = 114000$

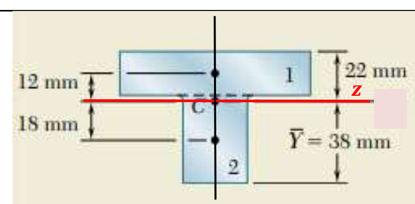
$$\bar{y} = \frac{\Sigma A_i y_i}{\Sigma A_i} = 38\ mm$$



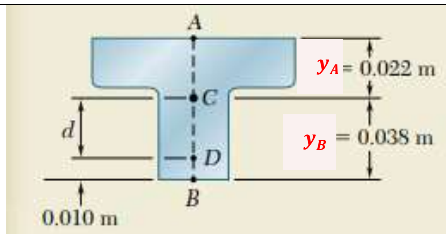
Moment of inertia of the cross-section about the neutral axis (centroidal z-axis)

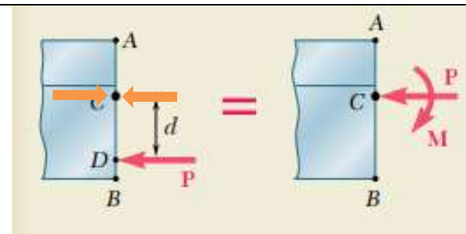
$$I_{zz} = \left\{\frac{90 \times 20^3}{12} + 1800 \times 12^2\right\} + \left\{\frac{30 \times 40^3}{12} + 1200 \times 18^2\right\}$$

$$I_{zz} = 868 \times 10^3\ mm^4 = 868 \times 10^{-9}\ m^4$$



The line of action of P is passing through point D in the cross section which is $d = 28mm$ below the centroid C. Transferring the force P from D to C by applying two equal and opposite forces of magnitude P at C, we can find the equivalent force-moment system at C.





The moment $M_z = P \times d = 0.028P \; clockwise = -0.028P$

Now, we can see that the effect of P acting eccentrically at D is equivalent to a axial compressive force P at C and pure bending moment $M_z = -0.028P$

The uniform normal stress on the cross section due to the axial compressive force P acting at the centroid,

$$\sigma_{xx}{}^{\text{axial}} = -\frac{P}{A} = -\frac{P}{3000\; mm^2} = -\frac{P}{3000 \times 10^{-6}\; m^2} = -333P$$

The normal stress on the cross section due to the bending moment $M_z = -0.028P$

$$\sigma_{xx}{}^{\text{bending}} = -\frac{M_z}{I_{zz}} y$$

The stress at B where $y_B = -0.038\; m$, $\sigma_{xx}{}^{\text{bending}} = -\frac{(-0.028P)}{868 \times 10^{-9}\; m^4}(-0.038\; m) = -1226P$

The stress at A where $y_A = +0.022\; m$, $\sigma_{xx}{}^{\text{bending}} = -\frac{(-0.028P)}{868 \times 10^{-9}\; m^4}(+0.022\; m) = +710P$

Applying the principle of superposition, we can find out the net stress acting on the cross section.

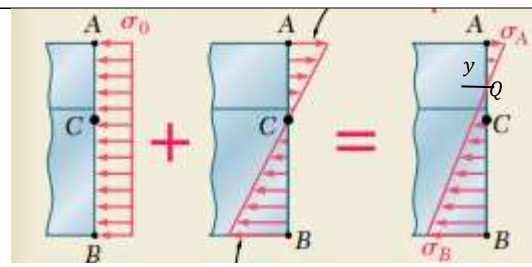
$$\sigma^A_{xx} = -333P + 710P = +377P$$
$$\sigma^B_{xx} = -333P - 1226P = -1559P$$

The maximum tensile stress permitted is $30MPa$.

$$30 \times 10^6 = 377P \ggg P = 79.6 \times 10^3 N$$

The maximum compressive stress permitted is $120MPa$.

$$-120 \times 10^6 = -1559P \ggg P = 77 \times 10^3 N$$



The magnitude of largest force P that can be applied without exceeding either of the allowable stress is the smaller of the two values of P found out.

$$P = 77 \times 10^3 N$$

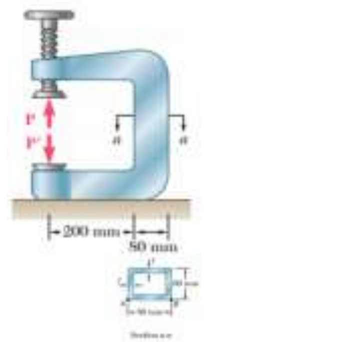
$$\sigma^A_{xx} = +377P = 29\; MPa \quad \lll\ggg \quad \sigma^B_{xx} = -1559P = 120\; MPa$$

Note the shift in the neutral axis position.

$$\frac{\sigma^A_{xx}}{y} = \frac{\sigma^B_{xx}}{0.06 - y} \ggg \frac{29}{y} = \frac{120}{0.06 - y} \ggg y = 0.0117\; m = 11.7\; mm$$

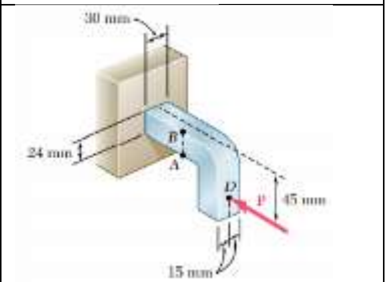
Exercise 1

The vertical portion of the press shown consists of a rectangular tube of wall thickness $t = 10\,mm$ and outer dimensions $80\,mm \times 60\,mm$. Knowing that the press has been tightened on wooden planks being glued together until $P = 20\,kN$, determine the stress at point A and point B.



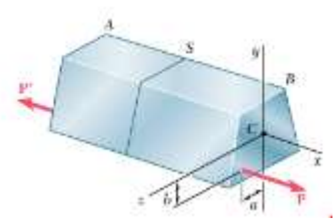
Exercise 2

Knowing that the magnitude of horizontal force is P=8 kN, determine the stresses at A and B.

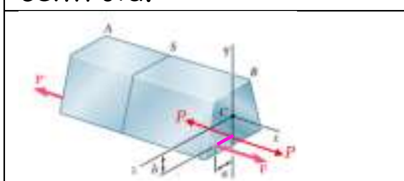


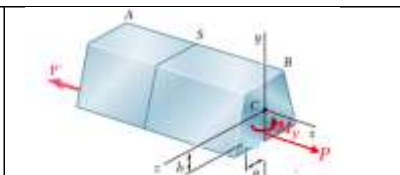
General case of eccentric axial loading

In the last section, we discussed about the eccentric axial loading in the plane of symmetry. Now, let us consider the analysis of axial eccentric loading not in the plane of symmetry. Consider a straight beam with a vertical axis of symmetry subjected to an axial force P not in the plane of symmetry.



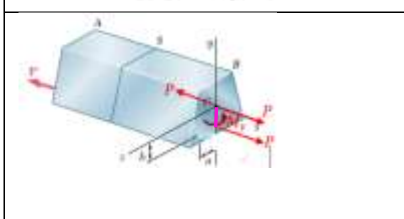
First, let us transfer the force P to the plane of symmetry and then to the centroid.

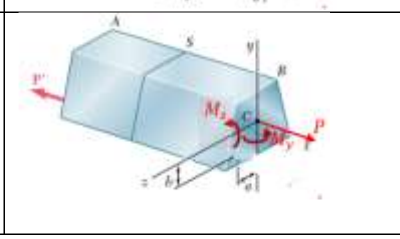


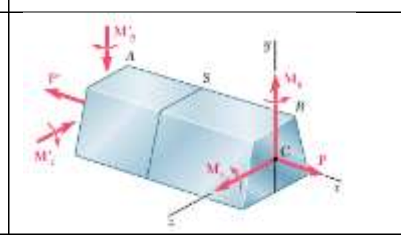


The moment $M_y = Pa$

The moment $M_z = Pb$







The net stress developed will be the algebraic sum of the stresses produced by the axial force P and the pure bending moments M_y and M_z (superposition).

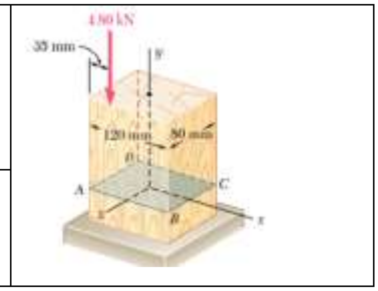
$$\sigma_{xx} = \pm\frac{P}{A} \pm \left(\frac{M_z}{I_{zz}}y\right) \pm \left(\frac{M_y}{I_{yy}}z\right)$$

#Example 1

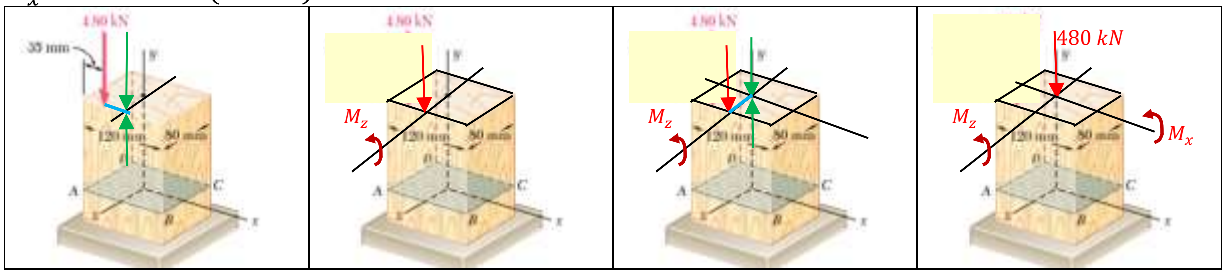
A vertical 480 kN load is applied as shown on a wooden post of rectangular cross section 80mm x 120 mm. Determine the stress at points A, B, C, and D.

$$I_{zz} = \frac{0.080 \times 0.120^3}{12} = 11.52 \times 10^{-6}\, m^4$$

$$I_{xx} = \frac{0.120 \times 0.08^3}{12} = 5.12 \times 10^{-6}\, m^4$$



To begin with, transfer the force P to the plane of symmetry. The force will be accompanied by a couple of moment $M_z = 480\, kN \times (60mm - 35mm) = 120\, N.m$. Then, transfer the force to the centroid of the cross section. The accompanying moment is $M_x = 480\, kN \times (40mm) = 192\, N.m$



The uniform normal stress produced over the cross section ABCD due to the axial force P,

$$\sigma_{yy}^{Axial} = -\frac{P}{A} = -\frac{480 \times 10^3\, N}{0.120\, m \times 0.080\, m} = -0.5 \times 10^6\, Pa$$

The normal stress on the cross section due to a moment $M_z = -120\, Nm$,

$$\sigma_{yy}^{bending} = \frac{-M_z}{I_{zz}} x$$

The normal stress at A, D due to the bending moment M_z, where $x = -0.06\, m$

$$\sigma_{yy}^{bending} = -\frac{-120\, N.m}{11.52 \times 10^{-6}\, m^4} \times (-0.06\, m) = -0.625 \times 10^6\, Pa$$

The normal stress at B, C due to the bending moment M_z, where $x = +0.06\, m$

$$\sigma_{yy}^{bending} = -\frac{-120\, N.m}{11.52 \times 10^{-6}\, m^4} \times (+0.06\, m) = +0.625 \times 10^6\, Pa$$

The normal stress on the cross section due to a moment $M_x = +192\, Nm$,

$$\sigma_{yy}^{bending} = \frac{-M_x}{I_{xx}} z$$

The normal stress at A, B due to the bending moment M_x, where $z = +0.04\, m$

$$\sigma_{yy}^{bending} = -\frac{192\, N.m}{5.12 \times 10^{-6}\, m^4} \times (+0.04\, m) = -1.5 \times 10^6\, Pa$$

The normal stress at C, D due to the bending moment M_x, where $z = -0.04\, m$

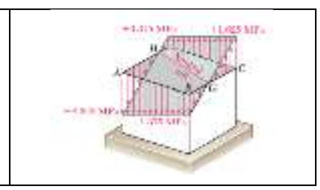
$$\sigma_{yy}^{bending} = -\frac{192\, N.m}{5.12 \times 10^{-6}\, m^4} \times (-0.04\, m) = +1.5 \times 10^6\, Pa$$

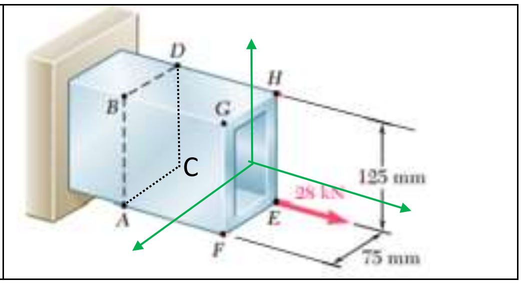
$$\sigma_{yy}^{A} = -0.5 \times 10^6\, Pa - 0.625 \times 10^6\, Pa - 1.5 \times 10^6\, Pa = -2.625 \times 10^6\, Pa$$

$$\sigma_{yy}^{B} = -0.5 \times 10^6\, Pa + 0.625 \times 10^6\, Pa - 1.5 \times 10^6\, Pa = -1.375 \times 10^6\, Pa$$

$$\sigma_{yy}^{C} = -0.5 \times 10^6\, Pa + 0.625 \times 10^6\, Pa + 1.5 \times 10^6\, Pa = +1.625 \times 10^6\, Pa$$

$$\sigma_{yy}^{D} = -0.5 \times 10^6\, Pa - 0.625 \times 10^6\, Pa + 1.5 \times 10^6\, Pa = +0.375 \times 10^6\, Pa$$



Exercise 1 The tube shown has a thickness 12 mm. Determine the stresses at points A, B, C, and D	

1. Mechanics of Materials, 2/e, J. M. Gere & S. P.Timoshenko, CBS Publishers.
2. Mechanics of Materials, 6/e, F.P. Beer & E. R. Johnston, McGraw Hill
3. Engineering Mechanics of Solids, 2/e, E. P. Popov, Pearson Education Asia.
4. Mechanics of Materials, 9/e, R. C. Hibbeler, Prentice Hall.